CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

SILVA DAIRY FARMS DAIRY & HEIFERS

1499 EDMINSTER RD & 1904 EDMINSTER RD

STEVINSON CA 95374

PREPARED BY:

CAS
CARDOSO AG SERVICES

MARIANN PEDROSO
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Silva Dairy Farms

Physical address of dairy:

Number and Street	City	County	Zip Code
1499 N Edminster RD	Stevinson	Merced	95374

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1915

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X055-X210-X049-XXXX

B. OPERATORS

Silva, Manuel

Operator name: Silva, Manuel

Telephone no.: Landline: ; Cellular: (209) 652-6582

Mailing Address Number and Street	City	State	Zip Code
1499 Edminster Rd	Stevinson	CA	95374

This operator is responsible for paying permit fees.

C. OWNERS

Silva, Manuel

Legal owner name: Silva, Manuel

Telephone no.: Landline: ; Cellular: (209) 652-6582

Mailing Address Number and Street	City	State	Zip Code
1499 Edminster Rd	Stevinson	CA	95374

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1,400	350	440	590	215	350
Maximum number	1,400	350	440	590	215	350
Average number	1,400	350	440	590	215	350
Avg live weight (lbs)	1,200	1,250	875	650		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 51,785.47 *tons per reporting period*

Total nitrogen from manure: 649,284.42 *lbs per reporting period*

After ammonia losses (30% loss applied): 454,499.09 *lbs per reporting period*

Total phosphorus from manure: 104,528.88 *lbs per reporting period*

Total potassium from manure: 263,513.92 *lbs per reporting period*

Total salt from manure: 739,672.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 14,202,000 *gallons*

Total nitrogen generated: 80,320.79 *lbs*

Total phosphorus generated: 16,065.58 *lbs*

Total potassium generated: 160,031.78 *lbs*

Total salt generated: 839,334.43 *lbs*

	14,202,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	14,202,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water
River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
3rd Street	38	38	2	manure	X055-X203-X005-XXXX X055-X203-X006-XXXX
Behind Heifers	13	13	2	both	X055-X210-X049-XXXX
Big Field	34	34	2	both	X055-X210-X032-XXXX X055-X210-X033-XXXX X055-X210-X047-XXXX
Field on the Right	36	36	2	process wastewater	X055-X210-X020-XXXX X055-X210-X021-XXXX
HR-1/HR-2/HR-3	81	81	2	both	X055-X210-X026-XXXX X055-X310-X018-XXXX X055-X310-X025-XXXX
HR-4	29	29	2	manure	X055-X210-X026-XXXX X055-X310-X016-XXXX X055-X310-X017-XXXX
JN-1	38	38	2	process wastewater	X055-X021-X019-XXXX X055-X310-X003-XXXX X055-X310-X004-XXXX
JN-3	19	19	2	process wastewater	X055-X310-X006-XXXX
JN-5	21	21	2	manure	X055-X210-X051-XXXX
Palma/Hinds	37	37	2	manure	X055-X210-X029-XXXX X055-X210-X030-XXXX
Pasture	20	18	0	manure	X055-X290-X010-XXXX
Totals for areas that were used for application	366	364	20		
Totals for areas that were not used for application					
Land application area totals	366	364	20		

B. CROPS AND HARVESTS

3rd Street

Field name: 3rd Street

3rd Street

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	652.10 *ton*	Dry-weight		80.4	29,600.00	4,300.00	45,100.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.16	199.12	28.93	303.38	948.50

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	781.80 *ton*	Dry-weight		62.6	11,300.00	2,400.00	14,800.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.57	173.90	36.93	227.76	800.23

Behind Heifers

Field name: Behind Heifers

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	211.00 *ton*	Dry-weight		77.5	21,600.00	2,300.00	29,800.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.23	157.76	16.80	217.65	715.78

Behind Heifers

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	153.70 *ton*	Dry-weight		62.6	11,300.00	2,400.00	14,800.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	11.82	99.93	21.22	130.89	459.87

Big Field

Field name: Big Field

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	712.00 *ton*	Dry-weight		71.3	14,700.00	3,600.00	28,500.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.94	176.70	43.27	342.58	1,502.53

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	803.60 *ton*	Dry-weight		71.8	13,300.00	2,700.00	14,800.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.64	177.29	35.99	197.29	653.18

Field on the Right

Field name: Field on the Right

10/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	466.10 *ton*	Dry-weight		76.3	24,300.00	3,200.00	38,700.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.95	149.13	19.64	237.50	693.48

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	752.90 *ton*	Dry-weight		68.4	13,900.00	2,100.00	15,600.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.91	183.72	27.76	206.19	753.40

HR-1/HR-2/HR-3

Field name: HR-1/HR-2/HR-3

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 81 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,478.30 *ton*	Dry-weight		74.7	24,600.00	3,850.00	37,250.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.25	227.18	35.55	344.00	1,089.71

HR-1/HR-2/HR-3

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 81 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	1,786.50 *ton*	Dry-weight		64.7	11,300.00	2,000.00	18,300.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.06	175.95	31.14	284.95	980.99

HR-4

Field name: HR-4

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	529.30 *ton*	Dry-weight		72.5	28,700.00	4,200.00	36,200.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.25	288.10	42.16	363.39	1,194.58

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	639.60 *ton*	Dry-weight		69.8	11,100.00	2,400.00	15,400.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.06	147.87	31.97	205.15	852.56

JN-1

Field name: JN-1

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	508.40 *ton*	Dry-weight		75.1	20,500.00	3,400.00	35,800.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.38	136.59	22.65	238.53	812.85

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	976.60 *ton*	Dry-weight		65.6	12,400.00	2,200.00	17,400.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.70	219.25	38.90	307.66	1,113.94

JN-3

Field name: JN-3

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	203.70 *ton*	Dry-weight		71.3	15,400.00	2,300.00	26,100.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.72	94.77	14.15	160.62	504.62

JN-3

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	366.70 *ton*	Dry-weight		63.4	12,000.00	2,100.00	16,000.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	19.30	169.53	29.67	226.04	819.40

JN-5

Field name: JN-5

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	383.00 *ton*	Dry-weight		71.3	15,400.00	2,300.00	26,100.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.24	161.22	24.08	273.23	858.43

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	348.30 *ton*	Dry-weight		64.3	12,300.00	2,200.00	17,200.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	16.59	145.66	26.05	203.69	698.69

Palma/Hinds

Field name: Palma/Hinds

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	752.40 *ton*	Dry-weight		73.5	14,500.00	2,700.00	22,400.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.34	156.28	29.10	241.42	991.54

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	601.50 *ton*	Dry-weight		63.2	11,800.00	2,100.00	15,400.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	16.26	141.19	25.13	184.26	646.11

Pasture

Field name: Pasture

01/01/2015: Pasture

Crop: Pasture Acres planted: 18 Plant date: 01/01/2015

No harvests entered for this crop.

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.00	60.00	2.00	2.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

3rd Street - 10/20/2022: Oats, silage-soft dough

Field name: 3rd Street

Crop: Oats, silage-soft dough — Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	224.39	28.37	123.80	0.00	988.00 *ton*
Application event totals		224.39	28.37	123.80	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.62	0.00	0.00	202.92	5,422,605.00 *gal*
Application event totals		7.62	0.00	0.00	202.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.00	0.00	0.00	213.11	5,694,969.00 *gal*
Application event totals		8.00	0.00	0.00	213.11	

3rd Street - 05/24/2023: Corn, silage

Field name: 3rd Street

Crop: Corn, silage — Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	138.17	13.68	41.72	930.24	456.00 *ton*
Application event totals		138.17	13.68	41.72	930.24	

3rd Street - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.76	0.00	0.00	153.28	4,096,041.00 *gal*
Application event totals		5.76	0.00	0.00	153.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.63	0.00	0.00	149.85	4,004,372.00 *gal*
Application event totals		5.63	0.00	0.00	149.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.85	0.00	0.00	288.99	7,722,717.00 *gal*
Application event totals		10.85	0.00	0.00	288.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.35	0.00	0.00	142.39	3,805,167.00 *gal*
Application event totals		5.35	0.00	0.00	142.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.93	0.00	0.00	290.90	7,773,840.00 *gal*
Application event totals		10.93	0.00	0.00	290.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.80	0.00	0.00	180.93	4,835,127.00 *gal*
Application event totals		6.80	0.00	0.00	180.93	

3rd Street - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.19	0.00	0.00	297.93	7,961,586.00 *gal*
Application event totals		11.19	0.00	0.00	297.93	

09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.86	0.00	0.00	182.53	4,877,876.00 *gal*
Application event totals		6.86	0.00	0.00	182.53	

Behind Heifers - 10/18/2022: Oats, silage-soft dough

Field name: Behind Heifers

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	98.90	11.53	106.00	854.12	198,000.00 *gal*
River	Surface water	1.39	0.00	0.00	124.79	2,160,000.00 *gal*
Application event totals		100.28	11.53	106.00	978.91	

04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	98.42	15.48	114.97	716.39	180,000.00 *gal*
Irrigation Well	Ground water	0.74	0.00	0.00	3,815.80	1,920,000.00 *gal*
Application event totals		99.16	15.48	114.97	4,532.18	

Behind Heifers - 05/27/2023: Corn, silage

Behind Heifers - 05/27/2023: Corn, silage

Field name: Behind Heifers

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.57	11.84	36.11	805.02	135.00 *ton*
Application event totals		119.57	11.84	36.11	805.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.92	0.00	0.00	4,769.75	2,400,000.00 *gal*
Application event totals		0.92	0.00	0.00	4,769.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.69	0.00	0.00	152.52	2,640,000.00 *gal*
Application event totals		1.69	0.00	0.00	152.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.88	0.00	0.00	4,531.26	2,280,000.00 *gal*
Application event totals		0.88	0.00	0.00	4,531.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	121.32	2,100,000.00 *gal*
Application event totals		1.35	0.00	0.00	121.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.74	0.00	0.00	3,815.80	1,920,000.00 *gal*
Application event totals		0.74	0.00	0.00	3,815.80	

Behind Heifers - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.83	0.00	0.00	4,292.77	2,160,000.00 *gal*
Application event totals		0.83	0.00	0.00	4,292.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.54	0.00	0.00	138.66	2,400,000.00 *gal*
Application event totals		1.54	0.00	0.00	138.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.54	0.00	0.00	138.66	2,400,000.00 *gal*
Application event totals		1.54	0.00	0.00	138.66	

Big Field - 10/17/2022: Oats, silage-soft dough

Field name: Big Field

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	61.28	6.36	10.38	0.00	275.00 *ton*
Application event totals		61.28	6.36	10.38	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	82.50	9.62	88.43	712.53	432,000.00 *gal*
River	Surface water	1.35	0.00	0.00	121.94	5,520,000.00 *gal*
Application event totals		83.86	9.62	88.43	834.46	

Big Field - 10/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	94.08	14.80	109.90	684.78	450,000.00 *gal*
Irrigation Well	Ground water	0.74	0.00	0.00	3,829.82	5,040,000.00 *gal*
Application event totals		94.82	14.80	109.90	4,514.61	

Big Field - 05/26/2023: Corn, silage

Field name: Big Field

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	90.32	14.21	105.50	657.39	432,000.00 *gal*
Irrigation Well	Ground water	0.92	0.00	0.00	4,741.69	6,240,000.00 *gal*
Application event totals		91.24	14.21	105.50	5,399.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.71	0.00	0.00	153.74	6,960,000.00 *gal*
Application event totals		1.71	0.00	0.00	153.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.87	0.00	0.00	4,513.72	5,940,000.00 *gal*
Application event totals		0.87	0.00	0.00	4,513.72	

Big Field - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	67.94	6.53	94.21	660.48	450,000.00 *gal*
River	Surface water	1.34	0.00	0.00	120.61	5,460,000.00 *gal*
Application event totals		69.28	6.53	94.21	781.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.73	0.00	0.00	3,784.23	4,980,000.00 *gal*
Application event totals		0.73	0.00	0.00	3,784.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.83	0.00	0.00	4,285.76	5,640,000.00 *gal*
Application event totals		0.83	0.00	0.00	4,285.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	70.65	6.79	97.98	686.90	468,000.00 *gal*
River	Surface water	1.53	0.00	0.00	137.84	6,240,000.00 *gal*
Application event totals		72.19	6.79	97.98	824.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.53	0.00	0.00	137.84	6,240,000.00 *gal*
Application event totals		1.53	0.00	0.00	137.84	

Field on the Right - 10/19/2022: Oats, silage-soft dough

Field name: Field on the Right

Crop: Oats, silage-soft dough

Plant date: 10/19/2022

Field on the Right - 10/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	103.89	12.11	111.36	897.25	576,000.00 *gal*
River	Surface water	1.34	0.00	0.00	120.17	5,760,000.00 *gal*
Application event totals		105.23	12.11	111.36	1,017.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	92.41	14.54	107.94	672.61	468,000.00 *gal*
Irrigation Well	Ground water	0.73	0.00	0.00	3,789.30	5,280,000.00 *gal*
Application event totals		93.14	14.54	107.94	4,461.90	

Field on the Right - 05/26/2023: Corn, silage

Field name: Field on the Right

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	99.52	15.66	116.25	724.35	504,000.00 *gal*
Irrigation Well	Ground water	0.90	0.00	0.00	4,650.50	6,480,000.00 *gal*
Application event totals		100.42	15.66	116.25	5,374.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.67	0.00	0.00	150.21	7,200,000.00 *gal*
Application event totals		1.67	0.00	0.00	150.21	

Field on the Right - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.86	0.00	0.00	4,435.20	6,180,000.00 *gal*
Application event totals		0.86	0.00	0.00	4,435.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	71.86	6.90	99.66	698.64	504,000.00 *gal*
River	Surface water	1.32	0.00	0.00	118.92	5,700,000.00 *gal*
Application event totals		73.18	6.90	99.66	817.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.73	0.00	0.00	3,746.24	5,220,000.00 *gal*
Application event totals		0.73	0.00	0.00	3,746.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.82	0.00	0.00	4,219.90	5,880,000.00 *gal*
Application event totals		0.82	0.00	0.00	4,219.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	69.30	6.66	96.10	673.69	486,000.00 *gal*
River	Surface water	1.50	0.00	0.00	135.19	6,480,000.00 *gal*
Application event totals		70.80	6.66	96.10	808.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.50	0.00	0.00	135.19	6,480,000.00 *gal*
Application event totals		1.50	0.00	0.00	135.19	

HR-1/HR-2/HR-3 - 10/15/2022: Oats, silage-soft dough

Field name: HR-1/HR-2/HR-3

Crop: Oats, silage-soft dough

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.87	15.16	66.13	0.00	1,125.00 *ton*
Application event totals		119.87	15.16	66.13	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	80.80	9.42	86.61	697.86	1,008,000.00 *gal*
Canal	Surface water	7.62	0.00	0.00	202.92	11,558,710.00 *gal*
Application event totals		88.43	9.42	86.61	900.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	94.78	14.91	110.71	689.85	1,080,000.00 *gal*
Canal	Surface water	8.00	0.00	0.00	213.11	12,139,276.00 *gal*
Application event totals		102.78	14.91	110.71	902.96	

HR-1/HR-2/HR-3 - 05/28/2023: Corn, silage

Field name: HR-1/HR-2/HR-3

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.76	0.00	0.00	153.28	8,731,035.00 *gal*
Application event totals		5.76	0.00	0.00	153.28	

HR-1/HR-2/HR-3 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	66.61	30.43	224.31	737.33	1,008,000.00 *gal*
Canal	Surface water	5.63	0.00	0.00	149.85	8,535,634.00 *gal*
Application event totals		72.24	30.43	224.31	887.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.85	0.00	0.00	288.99	16,461,580.00 *gal*
Application event totals		10.85	0.00	0.00	288.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	53.90	11.79	203.62	1,010.30	1,080,000.00 *gal*
Canal	Surface water	5.35	0.00	0.00	142.39	8,111,013.00 *gal*
Application event totals		59.25	11.79	203.62	1,152.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.93	0.00	0.00	290.90	16,570,554.00 *gal*
Application event totals		10.93	0.00	0.00	290.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.80	0.00	0.00	180.93	10,306,454.00 *gal*
Application event totals		6.80	0.00	0.00	180.93	

HR-1/HR-2/HR-3 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	53.90	11.79	203.62	1,010.30	1,080,000.00 *gal*
Canal	Surface water	11.19	0.00	0.00	297.93	16,970,750.00 *gal*
Application event totals		65.09	11.79	203.62	1,308.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.86	0.00	0.00	182.53	10,397,579.00 *gal*
Application event totals		6.86	0.00	0.00	182.53	

HR-4 - 10/16/2022: Oats, silage-soft dough

Field name: HR-4

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	301.77	38.15	166.49	0.00	1,014.00 *ton*
Application event totals		301.77	38.15	166.49	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.62	0.00	0.00	202.92	4,138,304.00 *gal*
Application event totals		7.62	0.00	0.00	202.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.00	0.00	0.00	213.11	4,346,161.00 *gal*
Application event totals		8.00	0.00	0.00	213.11	

HR-4 - 05/28/2023: Corn, silage

Field name: HR-4

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	132.84	17.12	74.98	867.89	348.00 *ton*
Application event totals		132.84	17.12	74.98	867.89	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.76	0.00	0.00	153.28	3,125,926.00 *gal*
Application event totals		5.76	0.00	0.00	153.28	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.63	0.00	0.00	149.85	3,055,968.00 *gal*
Application event totals		5.63	0.00	0.00	149.85	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.85	0.00	0.00	288.99	5,893,652.00 *gal*
Application event totals		10.85	0.00	0.00	288.99	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.35	0.00	0.00	142.39	2,903,943.00 *gal*
Application event totals		5.35	0.00	0.00	142.39	

HR-4 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.93	0.00	0.00	290.90	5,932,667.00 *gal*
Application event totals		10.93	0.00	0.00	290.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.80	0.00	0.00	180.93	3,689,965.00 *gal*
Application event totals		6.80	0.00	0.00	180.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.19	0.00	0.00	297.93	6,075,947.00 *gal*
Application event totals		11.19	0.00	0.00	297.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.86	0.00	0.00	182.53	3,722,590.00 *gal*
Application event totals		6.86	0.00	0.00	182.53	

JN-1 - 10/16/2022: Oats, silage-soft dough

Field name: JN-1

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	92.35	18.21	258.04	1,252.28	576,000.00 *gal*
Canal	Surface water	7.62	0.00	0.00	202.92	5,422,605.00 *gal*
Application event totals		99.97	18.21	258.04	1,455.19	

JN-1 - 10/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	70.99	32.43	239.07	785.84	504,000.00 *gal*
Canal	Surface water	8.00	0.00	0.00	213.11	5,694,969.00 *gal*
Application event totals		78.99	32.43	239.07	998.95	

JN-1 - 05/25/2023: Corn, silage

Field name: JN-1

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.76	0.00	0.00	153.28	4,096,041.00 *gal*
Application event totals		5.76	0.00	0.00	153.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	81.13	37.06	273.22	898.10	576,000.00 *gal*
Canal	Surface water	5.63	0.00	0.00	149.85	4,004,372.00 *gal*
Application event totals		86.76	37.06	273.22	1,047.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.85	0.00	0.00	288.99	7,722,717.00 *gal*
Application event totals		10.85	0.00	0.00	288.99	

JN-1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	77.81	7.48	107.90	756.43	576,000.00 *gal*
Canal	Surface water	5.35	0.00	0.00	142.39	3,805,167.00 *gal*
Application event totals		83.15	7.48	107.90	898.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	28.96	773,840.00 *gal*
Application event totals		1.09	0.00	0.00	28.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.80	0.00	0.00	180.93	4,835,127.00 *gal*
Application event totals		6.80	0.00	0.00	180.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	77.81	7.48	107.90	756.43	576,000.00 *gal*
Canal	Surface water	11.19	0.00	0.00	297.93	7,961,586.00 *gal*
Application event totals		89.00	7.48	107.90	1,054.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.86	0.00	0.00	182.53	4,877,876.00 *gal*
Application event totals		6.86	0.00	0.00	182.53	

JN-3 - 10/17/2022: Oats, silage-soft dough

Field name: JN-3

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

JN-3 - 10/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.62	0.00	0.00	202.92	2,711,302.00 *gal*
Application event totals		7.62	0.00	0.00	202.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	81.13	37.06	273.22	898.10	288,000.00 *gal*
Canal	Surface water	8.00	0.00	0.00	213.11	2,847,485.00 *gal*
Application event totals		89.14	37.06	273.22	1,111.21	

JN-3 - 05/25/2023: Corn, silage

Field name: JN-3

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.76	0.00	0.00	153.28	2,048,021.00 *gal*
Application event totals		5.76	0.00	0.00	153.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	65.92	30.11	221.99	729.70	234,000.00 *gal*
Canal	Surface water	5.63	0.00	0.00	149.85	2,002,186.00 *gal*
Application event totals		71.55	30.11	221.99	879.55	

JN-3 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.85	0.00	0.00	288.99	3,861,358.00 *gal*
Application event totals		10.85	0.00	0.00	288.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	49.78	10.89	188.08	933.20	234,000.00 *gal*
Canal	Surface water	5.35	0.00	0.00	142.39	1,902,583.00 *gal*
Application event totals		55.13	10.89	188.08	1,075.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.93	0.00	0.00	290.90	3,886,920.00 *gal*
Application event totals		10.93	0.00	0.00	290.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.80	0.00	0.00	180.93	2,417,563.00 *gal*
Application event totals		6.80	0.00	0.00	180.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	49.78	10.89	188.08	933.20	234,000.00 *gal*
Canal	Surface water	11.19	0.00	0.00	297.93	3,980,793.00 *gal*
Application event totals		60.97	10.89	188.08	1,231.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.86	0.00	0.00	182.53	2,438,938.00 *gal*
Application event totals		6.86	0.00	0.00	182.53	

JN-5 - 10/17/2022: Oats, silage-soft dough

Field name: JN-5

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	196.98	20.45	33.37	0.00	546.00 *ton*
Application event totals		196.98	20.45	33.37	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.62	0.00	0.00	202.92	2,996,703.00 *gal*
Application event totals		7.62	0.00	0.00	202.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.00	0.00	0.00	213.11	3,147,220.00 *gal*
Application event totals		8.00	0.00	0.00	213.11	

JN-5 - 05/25/2023: Corn, silage

Field name: JN-5

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	132.84	17.12	74.98	867.89	252.00 *ton*
Application event totals		132.84	17.12	74.98	867.89	

JN-5 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.76	0.00	0.00	153.28	2,263,602.00 *gal*
Application event totals		5.76	0.00	0.00	153.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.63	0.00	0.00	149.85	2,212,942.00 *gal*
Application event totals		5.63	0.00	0.00	149.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.85	0.00	0.00	288.99	4,267,817.00 *gal*
Application event totals		10.85	0.00	0.00	288.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.35	0.00	0.00	142.39	2,102,855.00 *gal*
Application event totals		5.35	0.00	0.00	142.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.93	0.00	0.00	290.90	4,296,069.00 *gal*
Application event totals		10.93	0.00	0.00	290.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.80	0.00	0.00	180.93	2,672,044.00 *gal*
Application event totals		6.80	0.00	0.00	180.93	

JN-5 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.19	0.00	0.00	297.93	4,399,824.00 *gal*
Application event totals		11.19	0.00	0.00	297.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.86	0.00	0.00	182.53	2,695,669.00 *gal*
Application event totals		6.86	0.00	0.00	182.53	

Palma/Hinds - 10/18/2022: Oats, silage-soft dough

Field name: Palma/Hinds

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	196.57	20.41	33.30	0.00	960.00 *ton*
Application event totals		196.57	20.41	33.30	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.62	0.00	0.00	202.92	5,279,905.00 *gal*
Application event totals		7.62	0.00	0.00	202.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.00	0.00	0.00	213.11	5,545,102.00 *gal*
Application event totals		8.00	0.00	0.00	213.11	

Palma/Hinds - 05/27/2023: Corn, silage

Field name: Palma/Hinds

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	127.16	16.39	71.77	830.75	425.00 *ton*
Application event totals		127.16	16.39	71.77	830.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.76	0.00	0.00	153.28	3,988,251.00 *gal*
Application event totals		5.76	0.00	0.00	153.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.63	0.00	0.00	149.85	3,898,993.00 *gal*
Application event totals		5.63	0.00	0.00	149.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.85	0.00	0.00	288.99	7,519,487.00 *gal*
Application event totals		10.85	0.00	0.00	288.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.35	0.00	0.00	142.39	3,705,031.00 *gal*
Application event totals		5.35	0.00	0.00	142.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.93	0.00	0.00	290.90	7,569,265.00 *gal*
Application event totals		10.93	0.00	0.00	290.90	

Palma/Hinds - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.80	0.00	0.00	180.93	4,707,886.00 *gal*
Application event totals		6.80	0.00	0.00	180.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.19	0.00	0.00	297.93	7,752,071.00 *gal*
Application event totals		11.19	0.00	0.00	297.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.86	0.00	0.00	182.53	4,749,511.00 *gal*
Application event totals		6.86	0.00	0.00	182.53	

Pasture - 01/01/2015: Pasture

Field name: Pasture

Crop: Pasture

Plant date: 01/01/2015

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	45.46	4.72	7.70	0.00	108.00 *ton*
Application event totals		45.46	4.72	7.70	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.62	0.00	0.00	202.92	2,568,602.00 *gal*
Application event totals		7.62	0.00	0.00	202.92	

Pasture - 01/01/2015: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.00	0.00	0.00	213.11	2,697,617.00 *gal*
Application event totals		8.00	0.00	0.00	213.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.76	0.00	0.00	153.28	1,940,230.00 *gal*
Application event totals		5.76	0.00	0.00	153.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.63	0.00	0.00	149.85	1,896,808.00 *gal*
Application event totals		5.63	0.00	0.00	149.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	16.20	0.00	0.00	431.38	5,460,576.00 *gal*
Application event totals		16.20	0.00	0.00	431.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	17.72	0.00	0.00	471.84	5,972,668.00 *gal*
Application event totals		17.72	0.00	0.00	471.84	

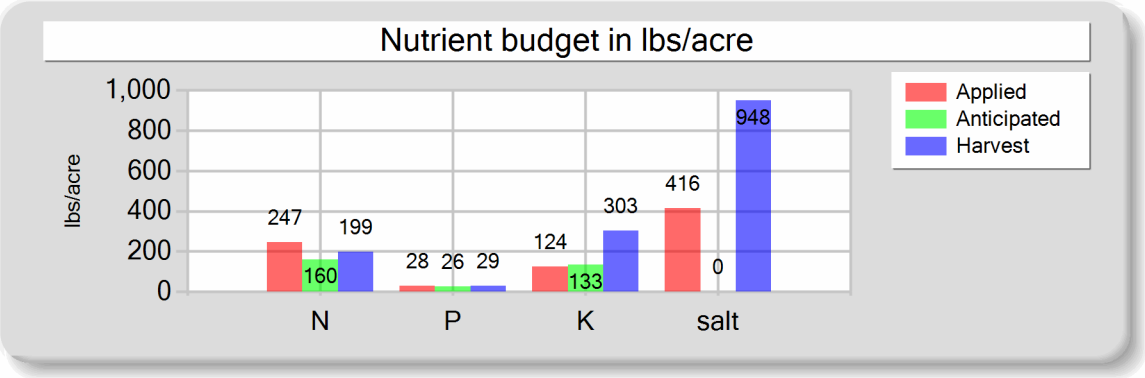
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	18.05	0.00	0.00	480.46	6,081,851.00 *gal*
Application event totals		18.05	0.00	0.00	480.46	

B. NUTRIENT BUDGET

3rd Street - 10/20/2022: Oats, silage-soft dough

Field name: 3rd Street | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	224.39	28.37	123.80	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.63	0.00	0.00	416.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.02	28.37	123.80	416.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	199.12	28.93	303.38	948.50
Nutrient balance	47.90	-0.55	-179.58	-532.47
Applied to removed ratio	1.24	0.98	0.41	0.44

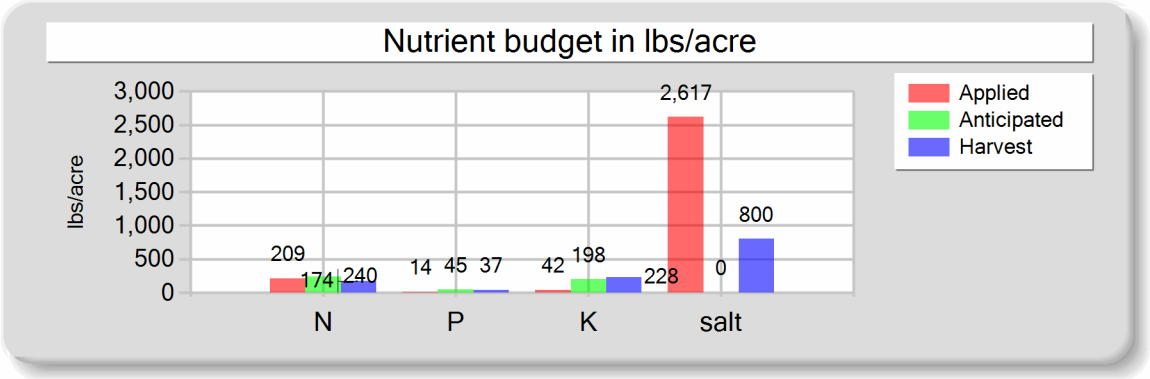
Fresh water applied	
11,117,574.00	*gallons*
409.42	*acre-inches*
10.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3rd Street - 05/24/2023: Corn, silage

Field name: 3rd Street | Crop: Corn, silage | Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	138.17	13.68	41.72	930.24
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	63.35	0.00	0.00	1,686.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.52	13.68	41.72	2,617.04
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	173.90	36.93	227.76	800.23
Nutrient balance	34.63	-23.25	-186.03	1,816.81
Applied to removed ratio	1.20	0.37	0.18	3.27

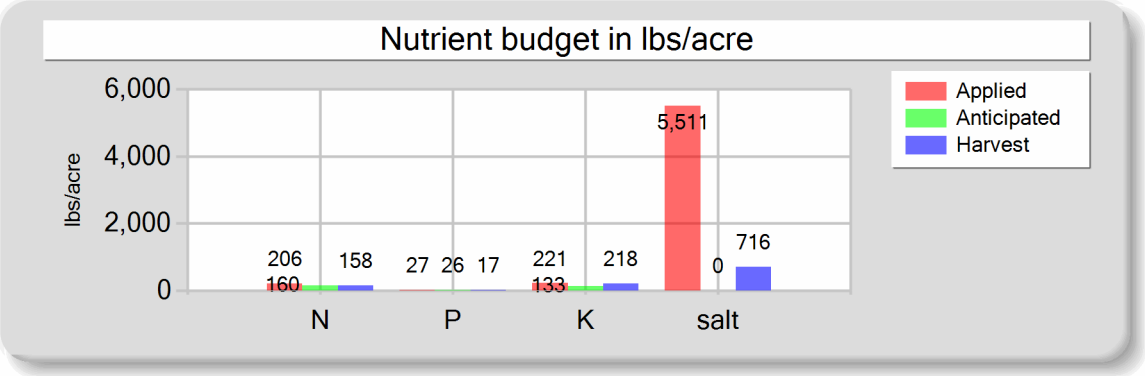
Fresh water applied	
45,076,726.00	*gallons*
1,660.02	*acre-inches*
43.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Behind Heifers - 10/18/2022: Oats, silage-soft dough

Field name: Behind Heifers Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.32	27.01	220.97	1,570.50
Fresh water	2.13	0.00	0.00	3,940.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.45	27.01	220.97	5,511.09
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	157.76	16.80	217.65	715.78
Nutrient balance	48.68	10.21	3.32	4,795.31
Applied to removed ratio	1.31	1.61	1.02	7.70

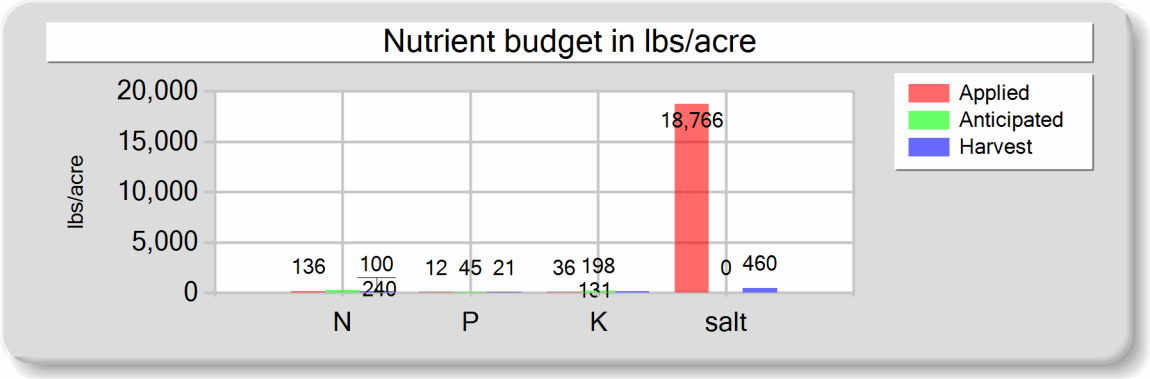
Fresh water applied	
4,080,000.00	*gallons*
150.25	*acre-inches*
11.56	*inches/acre*

Process wastewater applied	
378,000.00	*gallons*
13.92	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

Behind Heifers - 05/27/2023: Corn, silage

Field name: Behind Heifers Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	119.57	11.84	36.11	805.02
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.50	0.00	0.00	17,960.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	136.07	11.84	36.11	18,765.74
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	99.93	21.22	130.89	459.87
Nutrient balance	36.13	-9.39	-94.78	18,305.87
Applied to removed ratio	1.36	0.56	0.28	40.81

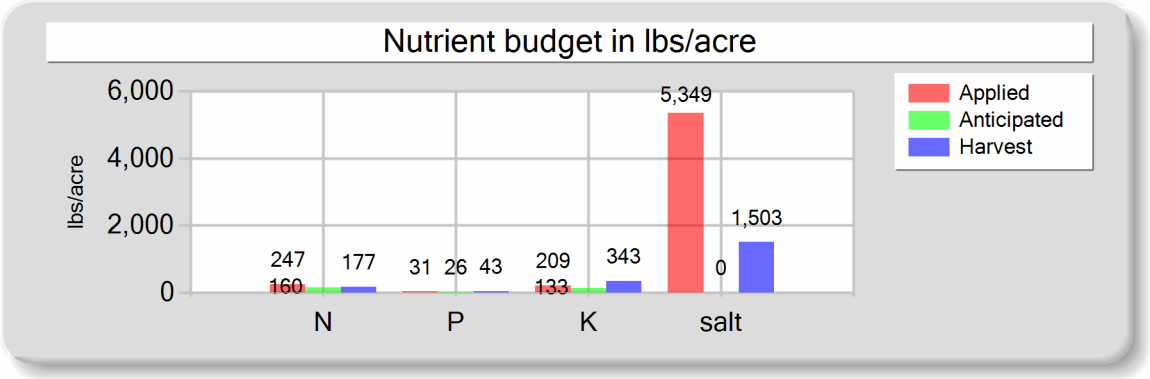
Fresh water applied	
18,300,000.00	*gallons*
673.93	*acre-inches*
51.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Big Field - 10/17/2022: Oats, silage-soft dough

Field name: Big Field | Crop: Oats, silage-soft dough | Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	61.28	6.36	10.38	0.00
Process wastewater	176.58	24.42	198.33	1,397.31
Fresh water	2.10	0.00	0.00	3,951.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.96	30.78	208.71	5,349.07
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	176.70	43.27	342.58	1,502.53
Nutrient balance	70.26	-12.49	-133.87	3,846.54
Applied to removed ratio	1.40	0.71	0.61	3.56

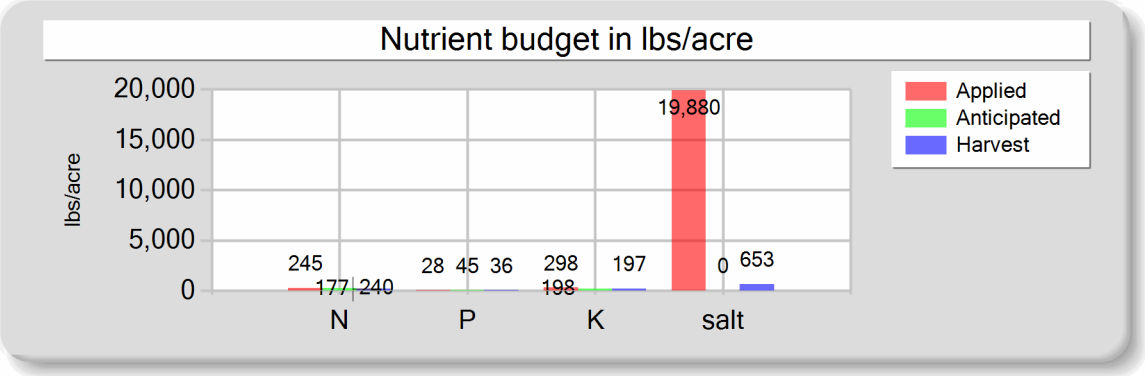
Fresh water applied	
10,560,000.00	*gallons*
388.89	*acre-inches*
11.44	*inches/acre*

Process wastewater applied	
882,000.00	*gallons*
32.48	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Big Field - 05/26/2023: Corn, silage

Field name: Big Field | Crop: Corn, silage | Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.91	27.52	297.69	2,004.77
Fresh water	9.47	0.00	0.00	17,875.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.38	27.52	297.69	19,880.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	177.29	35.99	197.29	653.18
Nutrient balance	68.08	-8.47	100.41	19,227.02
Applied to removed ratio	1.38	0.76	1.51	30.44

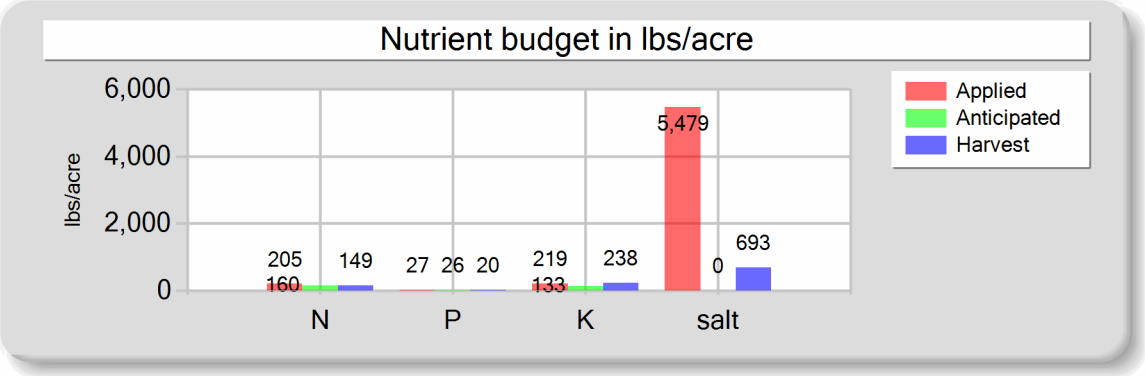
Fresh water applied	
47,700,000.00	*gallons*
1,756.63	*acre-inches*
51.67	*inches/acre*

Process wastewater applied	
1,350,000.00	*gallons*
49.72	*acre-inches*
1.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field on the Right - 10/19/2022: Oats, silage-soft dough

Field name: Field on the Right Crop: Oats, silage-soft dough Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.30	26.65	219.30	1,569.86
Fresh water	2.07	0.00	0.00	3,909.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.37	26.65	219.30	5,479.33
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	149.13	19.64	237.50	693.48
Nutrient balance	56.24	7.01	-18.20	4,785.85
Applied to removed ratio	1.38	1.36	0.92	7.90

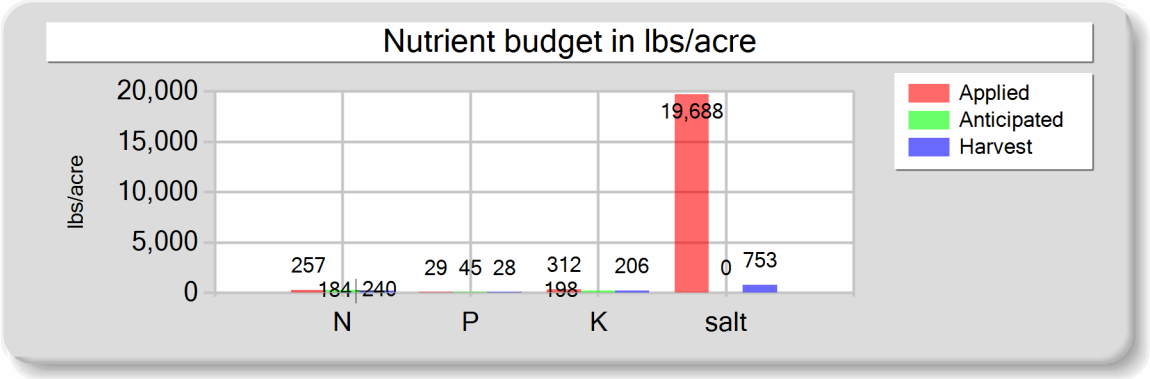
Fresh water applied	
11,040,000.00	*gallons*
406.57	*acre-inches*
11.29	*inches/acre*

Process wastewater applied	
1,044,000.00	*gallons*
38.45	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field on the Right - 05/26/2023: Corn, silage

Field name: Field on the Right Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.67	29.22	312.00	2,096.68
Fresh water	9.30	0.00	0.00	17,591.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	256.97	29.22	312.00	19,688.02
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	183.72	27.76	206.19	753.40
Nutrient balance	73.25	1.46	105.80	18,934.62
Applied to removed ratio	1.40	1.05	1.51	26.13

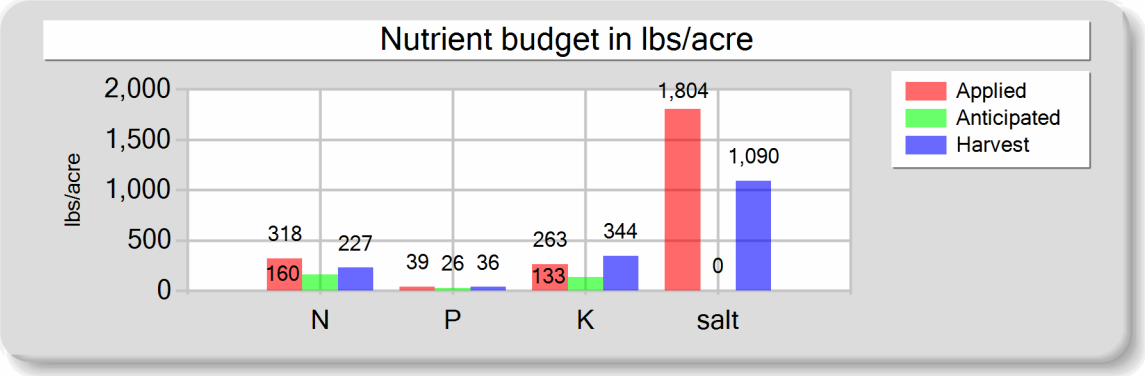
Fresh water applied	
49,620,000.00	*gallons*
1,827.34	*acre-inches*
50.76	*inches/acre*

Process wastewater applied	
1,494,000.00	*gallons*
55.02	*acre-inches*
1.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

HR-1/HR-2/HR-3 - 10/15/2022: Oats, silage-soft dough

Field name: HR-1/HR-2/HR-3 Crop: Oats, silage-soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	119.87	15.16	66.13	0.00
Process wastewater	175.58	24.33	197.32	1,387.72
Fresh water	15.63	0.00	0.00	416.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	318.07	39.48	263.45	1,803.75
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	227.18	35.55	344.00	1,089.71
Nutrient balance	90.90	3.93	-80.54	714.04
Applied to removed ratio	1.40	1.11	0.77	1.66

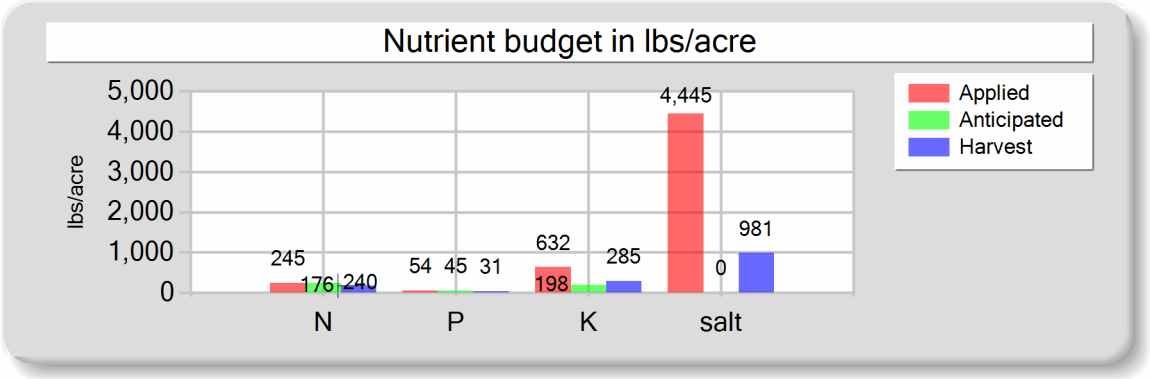
Fresh water applied	
23,697,986.00	*gallons*
872.72	*acre-inches*
10.77	*inches/acre*

Process wastewater applied	
2,088,000.00	*gallons*
76.89	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

HR-1/HR-2/HR-3 - 05/28/2023: Corn, silage

Field name: HR-1/HR-2/HR-3 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	174.40	54.02	631.55	2,757.93
Fresh water	63.35	0.00	0.00	1,686.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.76	54.02	631.55	4,444.73
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	175.95	31.14	284.95	980.99
Nutrient balance	68.80	22.87	346.60	3,463.75
Applied to removed ratio	1.39	1.73	2.22	4.53

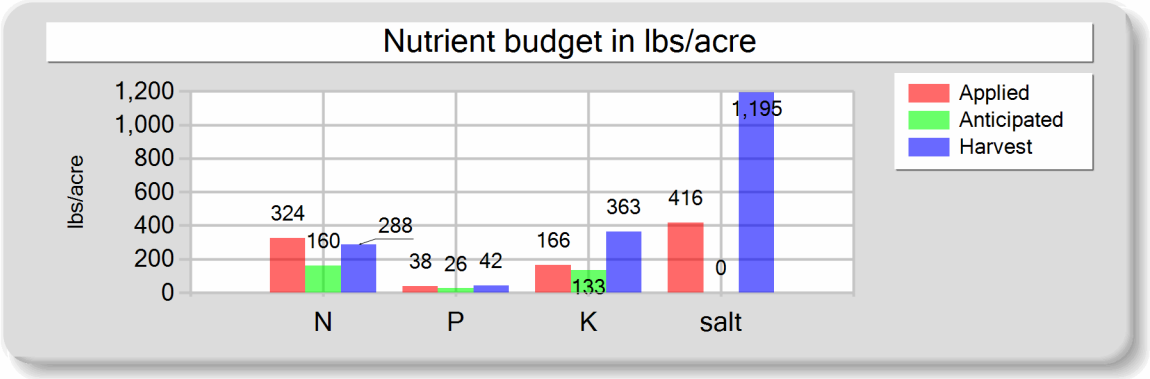
Fresh water applied	
96,084,599.00	*gallons*
3,538.47	*acre-inches*
43.68	*inches/acre*

Process wastewater applied	
3,168,000.00	*gallons*
116.67	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

HR-4 - 10/16/2022: Oats, silage-soft dough

Field name: HR-4 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	301.77	38.15	166.49	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.63	0.00	0.00	416.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	324.39	38.15	166.49	416.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	288.10	42.16	363.39	1,194.58
Nutrient balance	36.29	-4.01	-196.90	-778.55
Applied to removed ratio	1.13	0.90	0.46	0.35

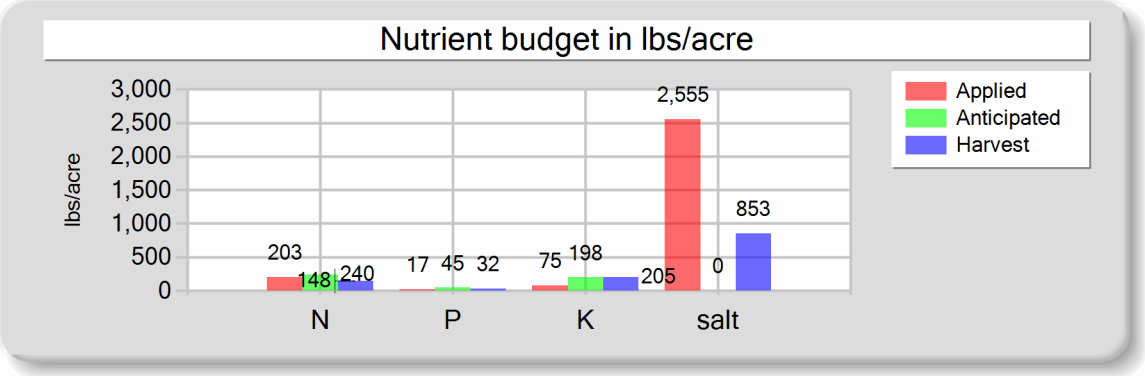
Fresh water applied	
8,484,465.00	*gallons*
312.45	*acre-inches*
10.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

HR-4 - 05/28/2023: Corn, silage

Field name: HR-4 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	132.84	17.12	74.98	867.89
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	63.35	0.00	0.00	1,686.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.19	17.12	74.98	2,554.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	147.87	31.97	205.15	852.56
Nutrient balance	55.33	-14.85	-130.17	1,702.13
Applied to removed ratio	1.37	0.54	0.37	3.00

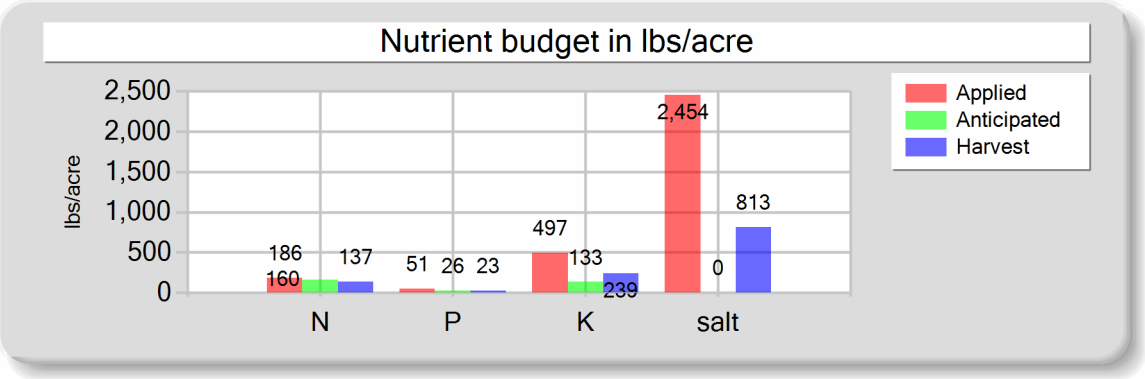
Fresh water applied	
34,400,658.00	*gallons*
1,266.86	*acre-inches*
43.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JN-1 - 10/16/2022: Oats, silage-soft dough

Field name: JN-1 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.34	50.64	497.12	2,038.11
Fresh water	15.63	0.00	0.00	416.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.97	50.64	497.12	2,454.14
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	136.59	22.65	238.53	812.85
Nutrient balance	49.38	27.99	258.59	1,641.29
Applied to removed ratio	1.36	2.24	2.08	3.02

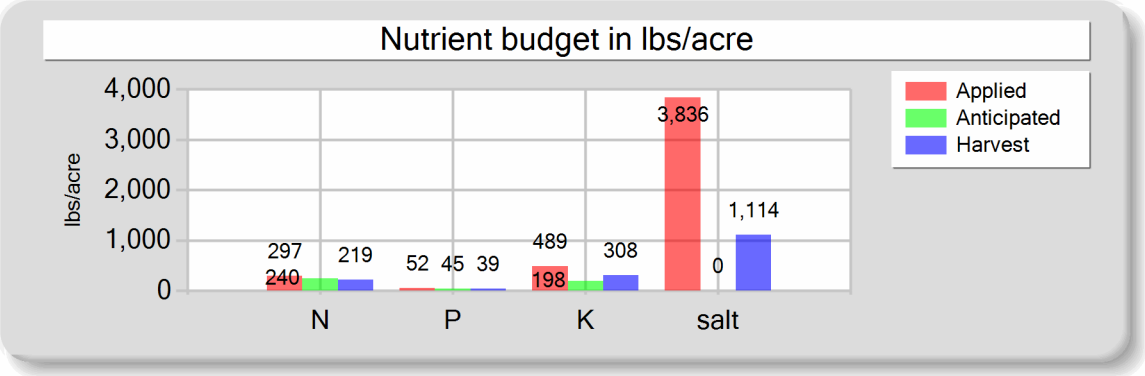
Fresh water applied	
11,117,574.00	*gallons*
409.42	*acre-inches*
10.77	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
1.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

JN-1 - 05/25/2023: Corn, silage

Field name: JN-1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	236.74	52.01	489.02	2,410.95
Fresh water	53.52	0.00	0.00	1,424.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.26	52.01	489.02	3,835.81
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	219.25	38.90	307.66	1,113.94
Nutrient balance	78.01	13.11	181.36	2,721.87
Applied to removed ratio	1.36	1.34	1.59	3.44

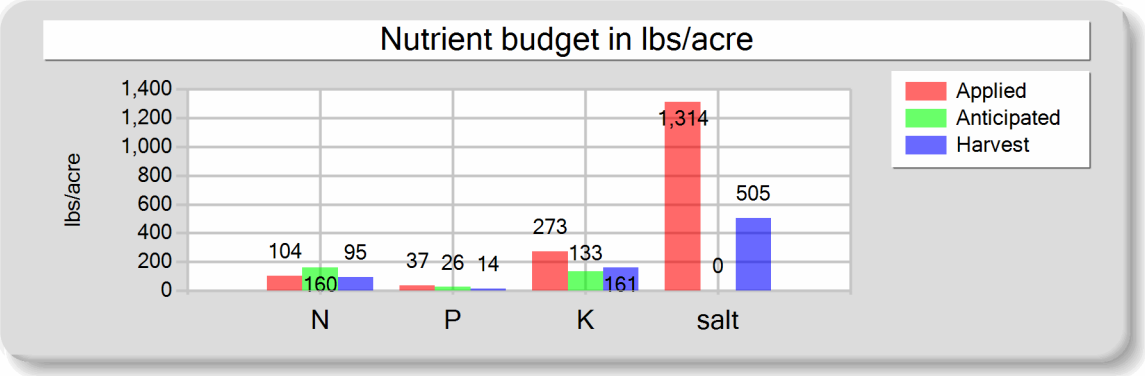
Fresh water applied	
38,076,726.00	*gallons*
1,402.24	*acre-inches*
36.90	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

JN-3 - 10/17/2022: Oats, silage-soft dough

Field name: JN-3 Crop: Oats, silage-soft dough Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	81.13	37.06	273.22	898.10
Fresh water	15.63	0.00	0.00	416.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	103.76	37.06	273.22	1,314.13
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	94.77	14.15	160.62	504.62
Nutrient balance	8.99	22.91	112.61	809.51
Applied to removed ratio	1.09	2.62	1.70	2.60

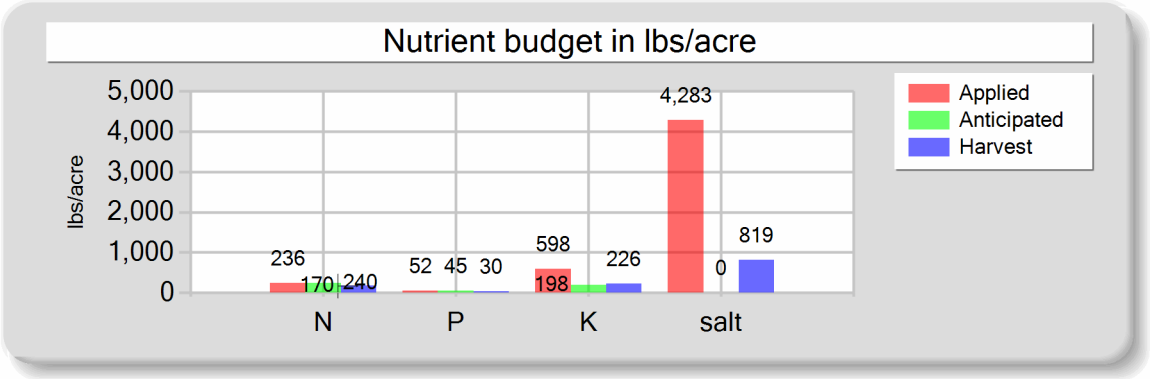
Fresh water applied	
5,558,787.00	*gallons*
204.71	*acre-inches*
10.77	*inches/acre*

Process wastewater applied	
288,000.00	*gallons*
10.61	*acre-inches*
0.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

JN-3 - 05/25/2023: Corn, silage

Field name: JN-3 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	165.49	51.90	598.15	2,596.10
Fresh water	63.35	0.00	0.00	1,686.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.84	51.90	598.15	4,282.91
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	169.53	29.67	226.04	819.40
Nutrient balance	66.31	22.23	372.11	3,463.51
Applied to removed ratio	1.39	1.75	2.65	5.23

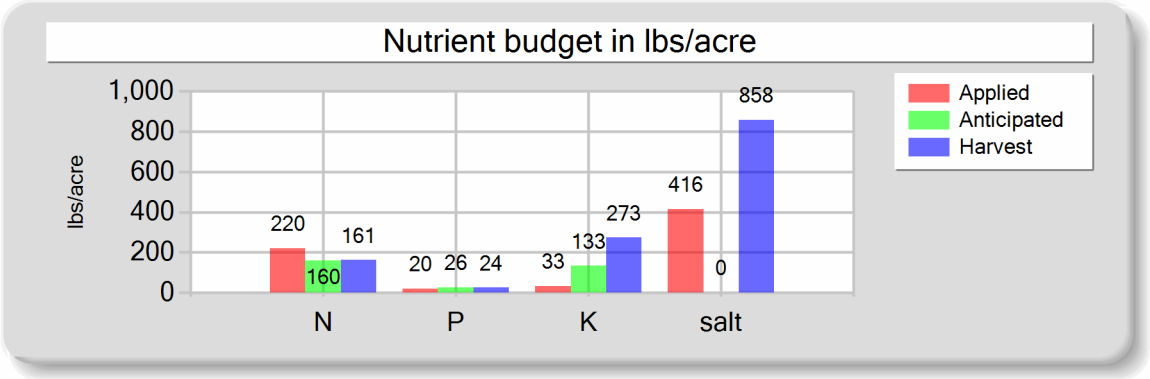
Fresh water applied	
22,538,362.00	*gallons*
830.01	*acre-inches*
43.68	*inches/acre*

Process wastewater applied	
702,000.00	*gallons*
25.85	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

JN-5 - 10/17/2022: Oats, silage-soft dough

Field name: JN-5 Crop: Oats, silage-soft dough Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	196.98	20.45	33.37	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.63	0.00	0.00	416.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.61	20.45	33.37	416.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	161.22	24.08	273.23	858.43
Nutrient balance	58.39	-3.63	-239.86	-442.40
Applied to removed ratio	1.36	0.85	0.12	0.48

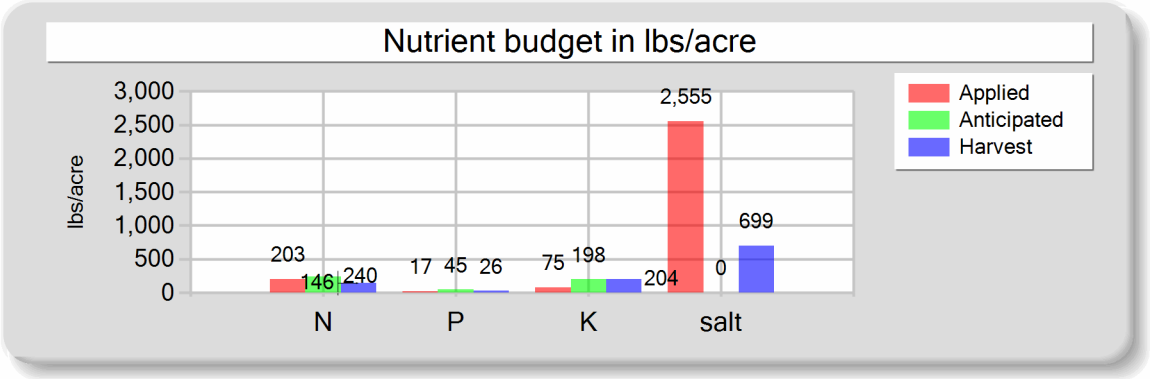
Fresh water applied	
6,143,923.00	*gallons*
226.26	*acre-inches*
10.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JN-5 - 05/25/2023: Corn, silage

Field name: JN-5 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	132.84	17.12	74.98	867.89
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	63.35	0.00	0.00	1,686.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.19	17.12	74.98	2,554.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	145.66	26.05	203.69	698.69
Nutrient balance	57.54	-8.93	-128.71	1,856.00
Applied to removed ratio	1.39	0.66	0.37	3.66

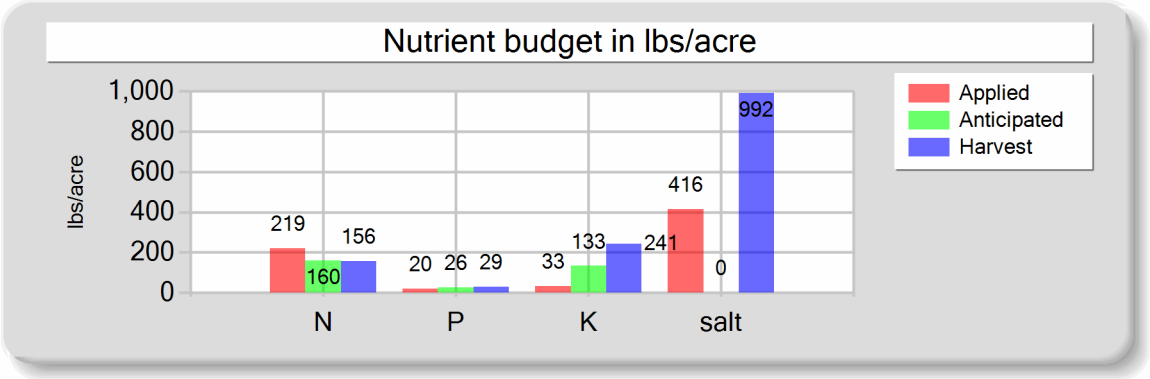
Fresh water applied	
24,910,822.00	*gallons*
917.38	*acre-inches*
43.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Palma/Hinds - 10/18/2022: Oats, silage-soft dough

Field name: Palma/Hinds Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	196.57	20.41	33.30	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.63	0.00	0.00	416.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.20	20.41	33.30	416.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	156.28	29.10	241.42	991.54
Nutrient balance	62.92	-8.69	-208.12	-575.51
Applied to removed ratio	1.40	0.70	0.14	0.42

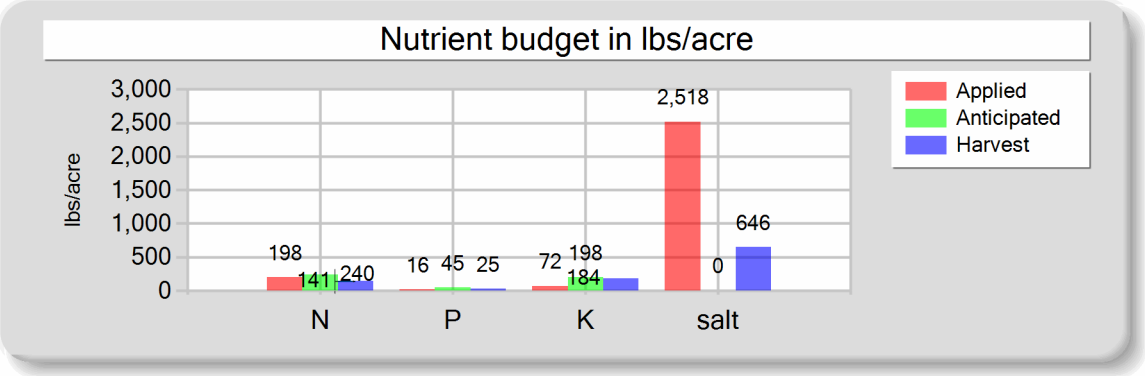
Fresh water applied	
10,825,007.00	*gallons*
398.65	*acre-inches*
10.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Palma/Hinds - 05/27/2023: Corn, silage

Field name: Palma/Hinds Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.16	16.39	71.77	830.75
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	63.35	0.00	0.00	1,686.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.51	16.39	71.77	2,517.55
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	141.19	25.13	184.26	646.11
Nutrient balance	56.32	-8.74	-112.49	1,871.44
Applied to removed ratio	1.40	0.65	0.39	3.90

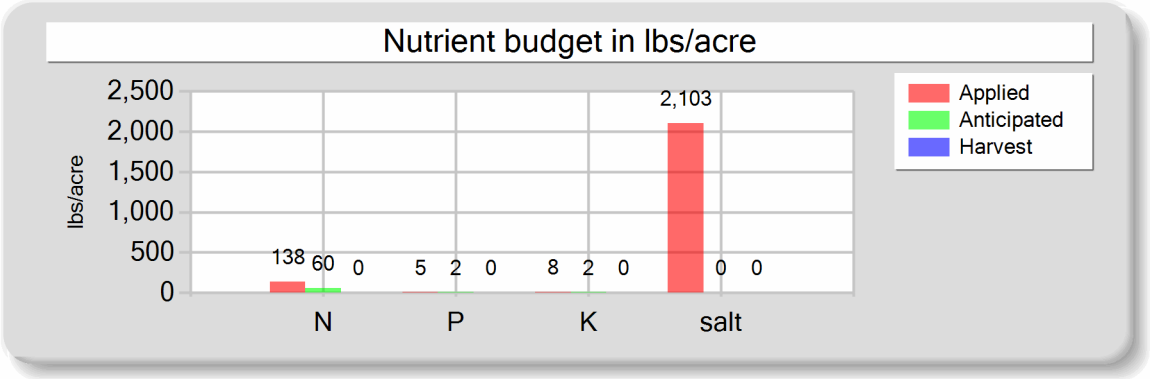
Fresh water applied	
43,890,495.00	*gallons*
1,616.34	*acre-inches*
43.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pasture - 01/01/2015: Pasture

Field name: Pasture Crop: Pasture Plant date: 01/01/2015



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	45.46	4.72	7.70	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	78.98	0.00	0.00	2,102.83
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	138.44	4.72	7.70	2,102.83
Anticipated crop nutrient removal	60.00	2.00	2.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	138.44	4.72	7.70	2,102.83
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied	
26,618,352.00	*gallons*
980.26	*acre-inches*
54.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Heifers Separator Manure

Sample and source description: 2022 Heifers Separator Manure

Sample date: 10/26/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,400.00	2,200.00	9,600.00							0.00
DL	100.00	100.00	100.00							1.00

2022 Separator Manure

Sample and source description: 2022 Separator Manure

Sample date: 10/26/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	1,900.00	3,100.00							0.00
DL	1.00	1.00	1.00							1.00

Dairy - Separator Manure

Sample and source description: Dairy - Separator Manure

Sample date: 05/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	2,000.00	6,100.00	10,000.00	3,200.00	4,400.00	3,500.00	3,000.00		13.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		1.00

Heifer - Separator Manure

Sample and source description: Heifer - Separator Manure

Sample date: 05/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,500.00	2,900.00	12,700.00	10,300.00	2,800.00	5,600.00	4,000.00	7,000.00		14.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		1.00

Dairy Corral Manure

Sample and source description: Dairy Corral Manure

Sample date: 10/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,600.00	2,700.00	11,400.00							0.00
DL	1.00	1.00	1.00							1.00

Heifers Corral Manure

Sample and source description: Heifers Corral Manure

Sample date: 10/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	4,700.00	28,700.00							0.00
DL	1.00	1.00	1.00							1.00

B. PROCESS WASTEWATER ANALYSES

2022 Lagoon Q4 Dairy

Sample and source description: 2022 Lagoon Q4 Dairy

Sample date: 12/06/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	776.00	413.00	0.00	2.10	90.70	834.00								11,600.00	6,720
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

2022 Lagoon Q4 heifers

Sample and source description: 2022 Lagoon Q4 heifers

Sample date: 12/06/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	728.00	284.00	0.00	2.10	144.00	2,040.00								15,800.00	9,900
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Dairy Lagoon Q1

Sample and source description: Dairy Lagoon Q1

Sample date: 01/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	173.00	134.00	0.00	0.80	42.90	311.00								3,900.00	2,320
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Heifers Lagoon Q1

Sample and source description: Heifers Lagoon Q1

Sample date: 03/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	642.00	263.00	0.00	1.50	197.00	1,580.00								12,500.00	7,300
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Dairy Lagoon Q2

Sample and source description: Dairy Lagoon Q2

Sample date: 05/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	851.00	432.00	0.00	0.80	134.00	995.00								13,000.00	6,200
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Heifer Lagoon Q2

Sample and source description: Heifer Lagoon Q2

Sample date: 05/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	640.00	289.00	0.00	1.40	293.00	2,160.00								15,100.00	7,100
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Dairy Lagoon Q3

Sample and source description: Dairy Lagoon Q3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	614.00	425.00	0.00	1.10	59.10	853.00								11,200.00	5,980
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Heifers Lagoon Q3

Sample and source description: Heifers Lagoon Q3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	483.00	260.00	0.00	1.40	106.00	1,830.00								14,700.00	9,080
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Dairy Lagoon Q4

Sample and source description: Dairy Lagoon Q4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	829.00	554.00	0.00	0.90	123.00	1,370.00								14,100.00	8,600
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Heifers Lagoon Q4

Sample and source description: Heifers Lagoon Q4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	769.00	575.00	0.00	0.90	124.00	158.00								15,200.00	10,200
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

C. FRESH WATER ANALYSES

Canal

Merquin Canal

Sample description: Merquin Canal

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.40	0.00	6.40								284.00	
DL	0.01	0.01	0.01								0.01	

Domestic Well

Domestic Well

DW Dairy

Sample description: DW Dairy

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.90	0.00	3.90	61.00	20.40	450.00	533.00	0.00	178.00	548.00	3,070.00	1,760
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1.00	1

DW Heifers

Sample description: DW Heifers

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	1.00	69.90	14.90	444.00	485.00	0.00	171.00	587.00	3,120.00	1,760
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1.00	1

DW Office

Sample description: DW Office

Sample date: 12/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.80	0.00	5.80	67.40	24.10	434.00	524.00	0.00	184.00	617.00	3,180.00	1,790
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1.00	1

Irrigation Well

Irrigation Well

IW Dairy

Sample description: IW Dairy

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.60	0.00	0.60								5,160.00	
DL	0.01	0.01	0.01								1.00	

River

River

Sample description: River

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	0.20								120.00	90
DL	0.01	0.01	0.01								1.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

3rd Street - 10/20/2022: Oats, silage-soft dough

3rd Street - 10/20/2022: Oats, silage-soft dough

3rd Street Oats

Sample and source description: 3rd Street Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,600.00	4,300.00	45,100.00		14.10
DL	0.01	0.01	0.01		1.00

3rd Street - 05/24/2023: Corn, silage

3rd Street Corn

Sample and source description: 3rd Street Corn

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,400.00	14,800.00		5.20
DL	0.01	0.01	0.01		1.00

Behind Heifers - 10/18/2022: Oats, silage-soft dough

Behind Heifers Oats

Sample and source description: Behind Heifers Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,300.00	29,800.00		9.80
DL	0.01	0.01	0.01		1.00

Behind Heifers - 05/27/2023: Corn, silage

Behind Heifers Corn

Sample and source description: Behind Heifers Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,400.00	14,800.00		5.20
DL	0.01	0.01	0.01		1.00

Big Field - 10/17/2022: Oats, silage-soft dough

Big Field Oats

Sample and source description: Big Field Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	3,600.00	28,500.00		12.50
DL	0.01	0.01	0.01		1.00

Big Field - 05/26/2023: Corn, silage

Big Field Corn

Sample and source description: Big Field Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,700.00	14,800.00		4.90
DL	0.01	0.01	0.01		1.00

Field on the Right - 10/19/2022: Oats, silage-soft dough

FOTR Oats

Sample and source description: FOTR Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,200.00	38,700.00		11.30
DL	0.01	0.01	0.01		1.00

Field on the Right - 05/26/2023: Corn, silage

FOTR Corn

Sample and source description: FOTR Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	2,100.00	15,600.00		5.70
DL	0.01	0.01	0.01		1.00

HR-1/HR-2/HR-3 - 10/15/2022: Oats, silage-soft dough

HR 1/2/3 B Oats

Sample and source description: HR 1/2/3 B Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	3,900.00	38,200.00		12.00
DL	0.01	0.01	0.01		1.00

HR-1/HR-2/HR-3 - 10/15/2022: Oats, silage-soft dough

HR 1/2/3 A Oats

Sample and source description: HR 1/2/3 A Oats

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,600.00	3,800.00	36,300.00		11.60
DL	0.01	0.01	0.01		1.00

HR-1/HR-2/HR-3 - 05/28/2023: Corn, silage

HR 1/2/3 Corn

Sample and source description: HR 1/2/3 Corn

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,000.00	18,300.00		6.30
DL	0.01	0.01	0.01		1.00

HR-4 - 10/16/2022: Oats, silage-soft dough

HR 4 Oats

Sample and source description: HR 4 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,700.00	4,200.00	36,200.00		11.90
DL	0.01	0.01	0.01		1.00

HR-4 - 05/28/2023: Corn, silage

HR 4 Corn

Sample and source description: HR 4 Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	2,400.00	15,400.00		6.40
DL	0.01	0.01	0.01		1.00

JN-1 - 10/16/2022: Oats, silage-soft dough

JN 1 Oats

Sample and source description: JN 1 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,500.00	3,400.00	35,800.00		12.20
DL	0.01	0.01	0.01		1.00

JN-1 - 05/25/2023: Corn, silage

JN 1 Corn

Sample and source description: JN 1 Corn

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,200.00	17,400.00		6.30
DL	0.01	0.01	0.01		1.00

JN-3 - 10/17/2022: Oats, silage-soft dough

JN3/JN5 Oats

Sample and source description: JN3/JN5 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	2,300.00	26,100.00		8.20
DL	0.01	0.01	0.01		1.00

JN-3 - 05/25/2023: Corn, silage

JN 3 Corn

Sample and source description: JN 3 Corn

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,100.00	16,000.00		5.80
DL	0.01	0.01	0.01		1.00

JN-5 - 10/17/2022: Oats, silage-soft dough

JN3/JN5 Oats

Sample and source description: JN3/JN5 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	2,300.00	26,100.00		8.20
DL	0.01	0.01	0.01		1.00

JN-5 - 05/25/2023: Corn, silage

JN 5 Corn

Sample and source description: JN 5 Corn

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,200.00	17,200.00		5.90
DL	0.01	0.01	0.01		1.00

Palma/Hinds - 10/18/2022: Oats, silage-soft dough

Palma/Hinds Oats

Sample and source description: Palma/Hinds Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	2,700.00	22,400.00		9.20
DL	0.01	0.01	0.01		1.00

Palma/Hinds - 05/27/2023: Corn, silage

Palma/Hinds Corn

Sample and source description: Palma/Hinds Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,100.00	15,400.00		5.40
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

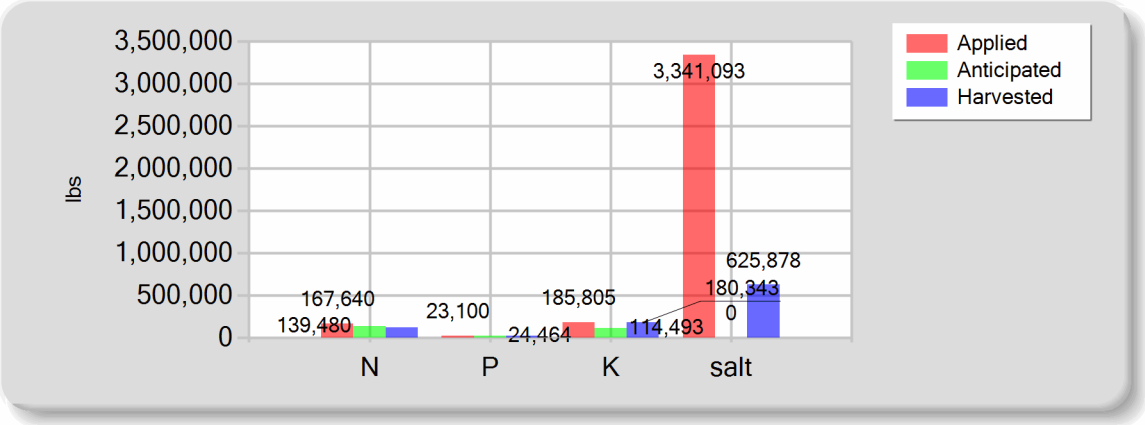
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

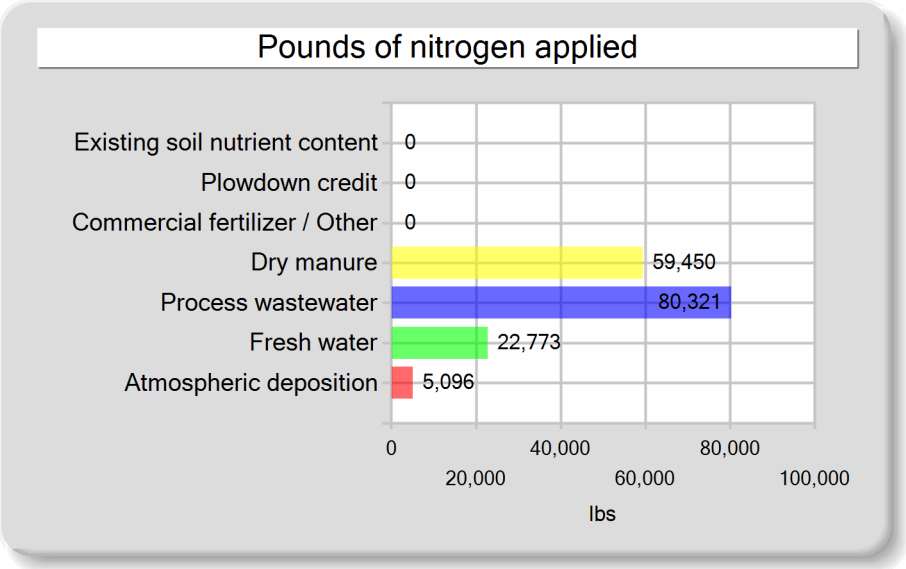
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

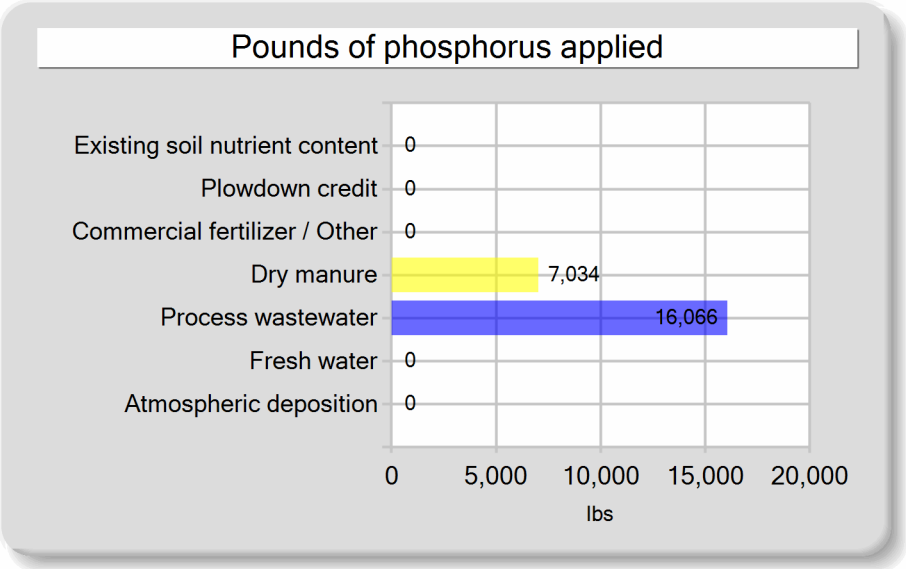
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	59,450.23	7,034.28	25,773.38	119,946.42
Process wastewater	80,320.79	16,065.58	160,031.78	839,334.43
Fresh water	22,773.05	0.00	0.00	2,381,812.60
Atmospheric deposition	5,096.00	0.00	0.00	0.00
Total nutrients applied	167,640.06	23,099.86	185,805.16	3,341,093.44
Anticipated crop nutrient removal	139,480.00	24,463.60	114,492.80	0.00
Actual crop nutrient removal	122,833.79	21,181.19	180,342.50	625,877.57
Nutrient balance	44,806.27	1,918.66	5,462.66	2,715,215.87
Applied to removed ratio	1.36	1.09	1.03	5.34

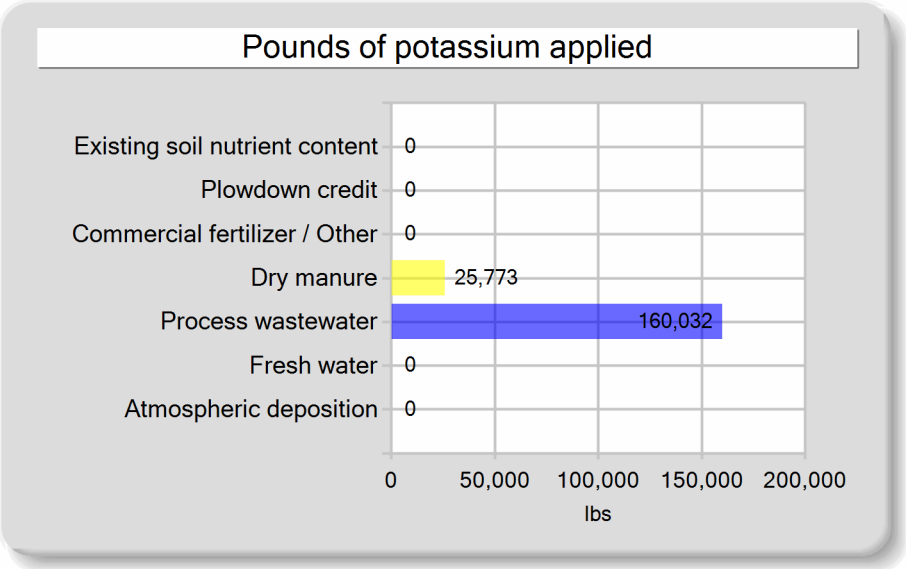
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

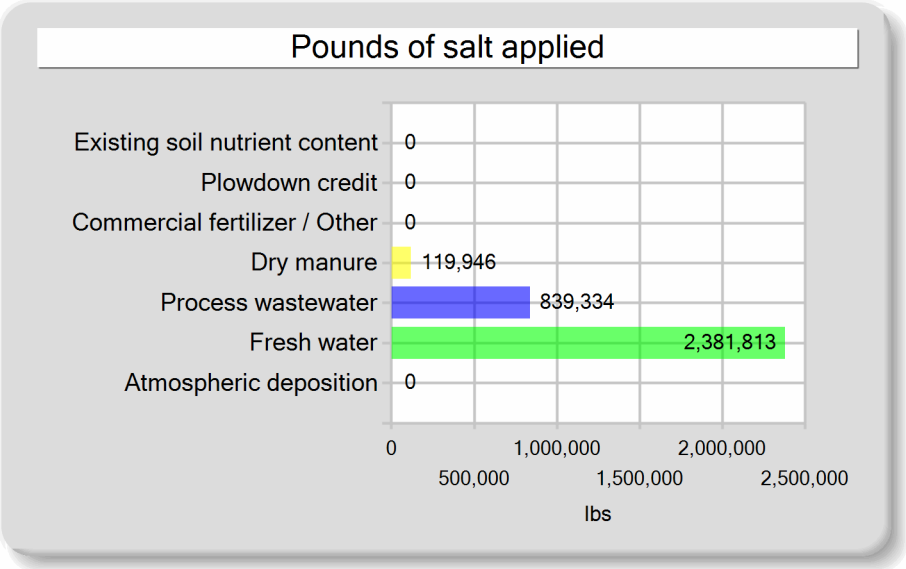


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

ANNUAL REPORT VALIDATION INFORMATION

A. VALIDATION ERRORS

The following sections contain validation errors and should be reviewed before submitting the Annual Report :

1. Harvest Events

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Manuel Silva

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuel Silva	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 1, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23G2065
Sampled By: M Pedroso
Requested By: Manuel Silva
Submitted Date: 07/27/23
Reported Date: 08/07/23
Project:
Crop ID:

Silva Dairy Farms
1499 Edminister Road
Stevinson CA 95375
00-0021334 08

E-mail: cas.labs@yahoo.com
Copy To:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	1W Dairy	7/26/23	10:21	5160	5.16	0.6	*		7.7

***** = Field NH_4-N not Taken.

ND = None Detected

Approved By: Scott M Friedland

Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

07/27/23 16:00

23G2065

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

Bill To: Acct No. 21334 Cons. 8

No. of Samples 1 No. Bottles 1

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Purchase Order No. Results Needed By

Client **Silva Dairy Farms**
Address 1499 Edminister Road
City, State, Zip **Stevinson, CA 95375**
Phone Fax
Cell/Email

Copy to Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by Manuel Silva - (209) 652-6582

Ranch

Date sampled 7/26/23

Sampled by MPedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) 1 L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES

		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. IW Dairy	Sampled From:	7/26/23	10:21A		0.4
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MPedroso	CAS INC	7/26/23 10:21AM	7/27/23 11AM
Second	JLF	DVL	7-27-23 11:00	7-27-23 3:35
Third				
Fourth		DVL	7/27 16:00	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ ____ In
$ ____ Out
Date ____

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23L0701
Sampled By: M Pedroso
Requested By: Manuel Silva
Submitted Date: 12/12/23
Reported Date: 12/21/23
Project:
Crop ID:

Silva Dairy Farms
1499 Edminister Road
Stevinson CA 95375
00-0021334 08

E-mail: cas.labs@yahoo.com
Copy To:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TDS mg/L	CO_3 as $CaCO_3$ mg/L	HCO_3 as $CaCO_3$ mg/L	Cl mg/L	SO_4 mg/L	Ca mg/L	Mg mg/L	Na mg/L	pH at 25°C unit
1	DW - Office	12/11/23	11:00	3180	3.18	5.8	ND		1790	ND	524	617	184	67.4	24.1	434	7.8
2	DW - Heifers	12/11/23	11:05	3120	3.12	ND	ND		1760	ND	485	587	171	69.9	14.9	444	7.8
3	DW - Dairy	12/11/23	11:08	3070	3.07	3.9	ND		1760	ND	533	548	178	61.0	20.4	450	7.9

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

12/12/23 15:51 23L0701

WATER WORK REQUEST

Bill To: Acct No. 21334 Cons. 8

Purchase Order No. Results Needed By

Client: Silva Dairy Farms
Address: 1499 Edminister Road
City, State, Zip: Stevinson, CA 95375
Phone: Fax:
Cell/Email:

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Manuel Silva - (209) 652-6582

Ranch:

Date sampled: 12/11/23

Sampled by: MPedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of Samples 3 No. Bottles 3

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

( ) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) 1 L plastic, unpreserved (white)

(✓) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-Office	Sampled From:	12/11/23	11AM	Ø	1.1
2. DW-Heifers	Sampled From:	12/11/23	11:05AM	Ø	2.6
3. DW-Dairy	Sampled From:	12/11/23	11:08AM	Ø	0.9
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MPed	CAS Inc.	12/11/23 11AM	12/12/23 @9:30am
Second	JLF	DLV	12-12-23 9:30	12-12-23 2:01
Third				
Fourth		DLV	12/12/23 13:51	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a late damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ ____ In
$ ____ Out

Signature ____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020